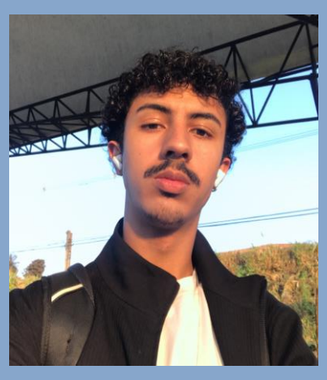

Opportunities and Challenges of Human-AI Interaction in Knowledge Work

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Introduction

This poster presents part of a qualitative research experience developed within a remote collaboration linked to Palacký University Olomouc. The broader study investigates the opportunities and challenges of human-AI interaction in knowledge work, with emphasis on productivity, innovative collaboration, ethical concerns, possible role transformation, and workforce well-being. According to the research documentation, the study was designed as a qualitative inquiry based on semi-structured in-depth interviews, allowing participants to expand beyond fixed questions and contribute their own relevant perceptions and experiences.

Goal

The goal of this work was to support the qualitative research process by understanding the scope of the project together with Professor Michal Müller, conducting and refining interviews with knowledge workers, and consolidating interview material for subsequent scientific analysis. More specifically, the work aimed to capture how participants perceive the use of generative AI in knowledge-intensive activities, especially regarding productivity, ethics, collaboration, professional roles, anxiety, and organizational adaptation.

Figure 1. *Qualitative research workflow for interview conduction and material consolidation*

Methodology / Discussed Topics

The methodological approach was based on semi-structured in-depth interviews. The interview guide explicitly encouraged open-ended exploration, follow-up questions, reformulation of similar questions when deeper answers were needed, and flexibility for the participant to shape the discussion. This is consistent with your description of the meetings with the professor and with the interview instructions contained in the guide

The main discussed topics included: participants' professional trajectories and responsibilities; prior experience with generative AI; AI use in daily knowledge work; suitable tasks for AI support or automation; benefits and limitations of AI; ethical concerns; human-AI collaboration; effects on professional roles; emerging skills and competencies; well-being, stress and adaptation; anxiety related to AI; recommendations for HR and organizational support; and future opportunities and challenges of AI in knowledge work.

Development

This work was developed through continuous alignment with the supervising researcher, semi-structured interview conduction, and consolidation of qualitative material for later scientific analysis. The interview process was not limited to the predefined script: questions were adapted, repeated, or reformulated whenever necessary to deepen relevant themes and capture richer participant reflections. Once completed, the interviews were recorded, transcribed, and organized as empirical material for the broader research project. The resulting material already points to recurring thematic patterns, especially regarding AI-supported productivity, ethical and governance concerns, the persistence of human criticality, and tensions between efficiency gains and anxiety in knowledge work.

Conclusion

This experience supported a broader qualitative study on human-AI interaction in knowledge work through interview preparation, conduction, and consolidation of empirical material for later scientific analysis. The interviews suggest that AI is perceived as both an opportunity and a challenge: it can improve productivity, accelerate analytical work, and expand possibilities for decision-making, but it also raises concerns related to reliability, privacy, bias, governance, and overdependence. Rather than making human workers obsolete, the material indicates that AI reinforces the importance of critical thinking, contextual judgment, and the ability to validate automated outputs. At the same time, the findings show that AI adoption is not only a technical issue, but also an ethical and organizational one, with direct implications for stress, anxiety, and well-being. Overall, the study points toward a future of knowledge work defined not by pure automation, but by a more demanding form of human-AI collaboration that requires literacy, governance, and responsible institutional support.